DO NOW — RECALL

Software development methods: evaluation

Name: _____ Class: _____ Date: _____

1. Independent tasks take 3 and 6 min; final integration 2 min. What is ideal parallel duration?

2. Name one XP practice intended to find code errors early.

3. Why would a user evaluate a prototype?

4. Simplify F=(¬A ∧ ¬B) ∨ (¬A ∧ B). State the law and final answer.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

max(3,6)+2=8 minutes.

2. Model answer

Test-driven development or continuous integration.

3. Model answer

To identify missing needs, usability issues or changes before more costly development.

4. Model answer

Factor ¬A: F=¬A∧(¬B∨B)=¬A∧1=¬A. This also forms a pair of adjacent 1s on a K-map.

Vocabulary — know these terms

evaluation — Judgement supported by evidence and limitations.

risk-driven — Development priorities guided by significant uncertainties.

iterative — Revisits improvements in cycles.

stakeholder — Person or organisation affected by a project.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check and explain

Qualify the recommendation: frequent prototyping must not replace security and validation testing.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

Avoid a universal answer

Method suitability depends on risk, requirements, regulation, user access and timescale.

Support with scenario evidence

A high-risk system may justify Spiral's explicit risk analysis despite extra overhead.

Include a drawback

Evaluate cost, documentation, flexibility and technical quality rather than merely naming a method.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

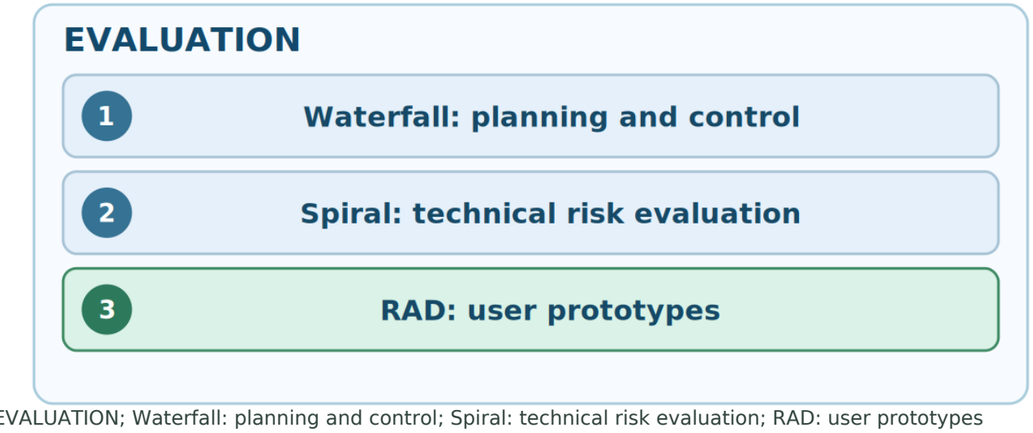
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



DECISION FACTORS; Are requirements stable?; Are risks high?; Is rapid user feedback available?

New Learning



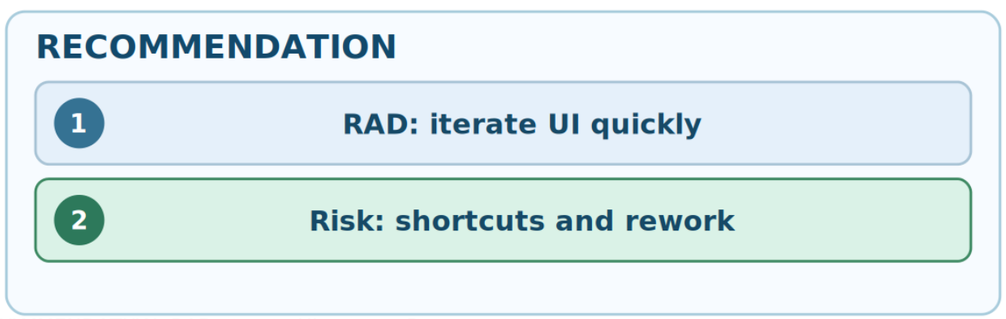
EVALUATION; Waterfall: planning and control; Spiral: technical risk evaluation; RAD: user prototypes

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A small charity wants a booking page within a month and volunteers can review mock-ups daily.

RAD fits frequent volunteer feedback and a requirement to develop the interface rapidly.



RECOMMENDATION; RAD: iterate UI quickly; Risk: shortcuts and rework

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

The best development methodology is the same for every software project.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

I DO — WORKED EXAMPLE

Follow the worked method

A safety-critical device uses unfamiliar hardware. Evaluate Spiral versus RAD.

Predict the method

Recommend Spiral because technical risks can be evaluated at each iteration.

Check and explain

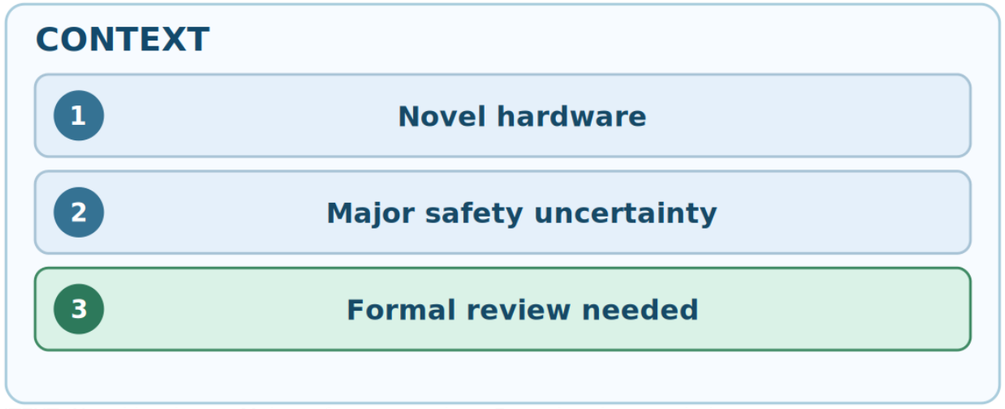
Explain the disadvantage: risk analysis adds project management and cost.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A safety-critical device uses unfamiliar hardware. Evaluate Spiral versus RAD.

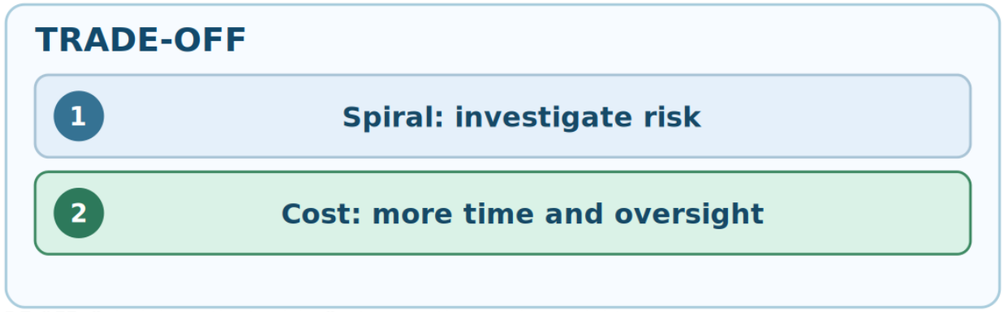
Recommend Spiral because technical risks can be evaluated at each iteration.



CONTEXT; Novel hardware; Major safety uncertainty; Formal review needed

A safety-critical device uses unfamiliar hardware. Evaluate Spiral versus RAD.

Recommend Spiral because technical risks can be evaluated at each iteration.



TRADE-OFF; Spiral: investigate risk; Cost: more time and oversight

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate Agile compared with Waterfall for a government reporting system where requirements are partly fixed but user interface needs may change.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give one defining feature each of Waterfall, Spiral, RAD and XP.

Question 2

Choose a development method for a regulated safety-critical control application with uncertain technology. Give three reasons and one drawback.

WE DO — GUIDED PRACTICE

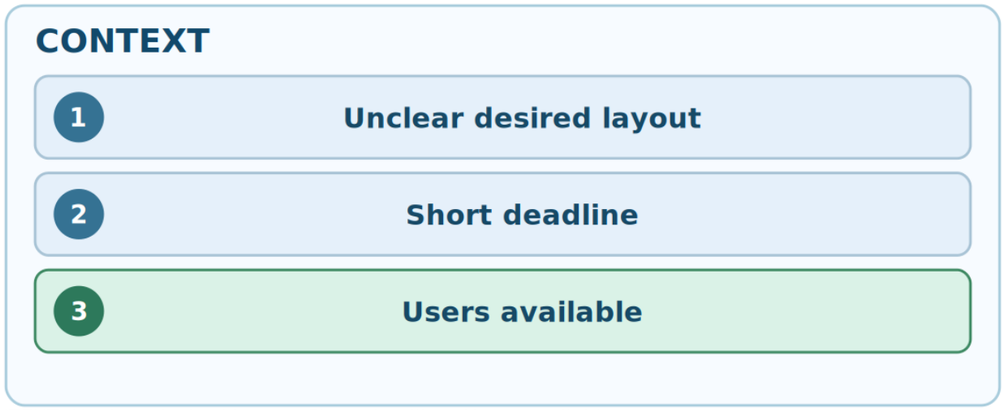
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A small charity wants a booking page within a month and volunteers can review mock-ups daily.

RAD fits frequent volunteer feedback and a requirement to develop the interface rapidly.



CONTEXT; Unclear desired layout; Short deadline; Users available

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict the method RAD fits frequent volunteer feedback and a requirement to develop the interface rapidly.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.